

Relational Geometry and the Emergence of Dimensions

A Minimal Generative Algebraic Model

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Status. This manuscript proposes an exploratory generative framework. It is not presented as a complete physical theory. Its aim is to define a minimal internal formalism, clearly separate postulates from derived relations, and distinguish algebraic results from programmatic physical correspondences.

Abstract

We define a minimal relational algebra in which new degrees of freedom are generated by successive projections relative to a distinguished direction. Starting from a normalized reference $\hat{N}$ and a differentiation marker R , with $\|R\| = R \in [0, 1]$, we postulate an antisymmetric bilinear operator $\times$ satisfying restricted norm compatibility, metric compatibility in the orthogonal subspace, and a double-projection closure rule with binary sign $\sigma \in \{\pm 1\}$. From these assumptions, a hierarchy

$$U, \quad V, \quad m, \quad a$$

emerges with controlled scaling in R . A quadratic harmonic depth ε is introduced through the closure condition

$$\frac{\omega_V^2}{\omega_I^2} = 2^{-\varepsilon}.$$

The baryonic instance $\varepsilon = 4$ is structurally motivated by the minimal real representation dimension associated with spin- $\frac{1}{2}$ behavior and yields a proton-scale consistency target. Further tentative correspondences, including atomic valence heuristics, a Hubble analogy, and a weak-field gravity limit, are presented as program notes rather than definitive derivations. The framework is explicitly exploratory: all physical correspondences are labeled as programmatic, conditional, and falsifiable targets.

Prologue: On the Spirit of This Inquiry

Many physical explanations are first learned as labels: light comes “by light”, magnets attract “by magnetism”, bodies fall “by gravity”. Labels predict, but they do not necessarily reveal what is being measured.

One question remains resistant to isolation: the present we inhabit. If time is defined by periodic comparison, perhaps it is not primitive, but a structure that appears once relations become stable enough to be compared.

The ancient Chinese philosopher Zhuangzi captured this journey from undifferentiated unity to fragmented multiplicity with remarkable precision:

El conocimiento de los antiguos era perfecto. ¿Cómo era perfecto? Al principio, todavía no sabían que había cosas. Este es el conocimiento más perfecto, porque no se le puede agregar nada. Luego, sabían que había cosas, pero no hacían distinciones entre ellas. Luego hicieron distinciones, pero aún no emitían juicios sobre ellas. Cuando surgieron los juicios, el Todo fue destruido. Con la destrucción del Todo, surgió el sesgo individual.

This work asks: what minimal algebraic structure must exist for this process of distinction and fragmentation to become coherent? A normalized reference $\hat{\mathbb{N}}$ encodes the primordial state where “there were not yet things”. A differentiation marker R introduces the first distinction. A generative operator $\times$ produces subsequent layers of differentiation until identity and orientation emerge.

1 Algebraic Foundations

1.1 Primitive elements

Let $\mathcal{V}$ be a real normed vector space.

Postulate 1 (Reference normalization).

$$\|\hat{\mathbf{R}}\| = 1, \quad \|R\| = R \in [0, 1].$$

Postulate 2 (Reference orthogonality).

$$\langle \hat{\mathbf{R}}, R \rangle = 0.$$

Define

$$V_R := \{A \in \mathcal{V} \mid \langle A, R \rangle = 0\}.$$

1.2 Generative operator

Postulate 3 (Bilinearity and antisymmetry). *For all $A, B, C \in \mathcal{V}$ and scalars α, β ,*

$$(\alpha A + \beta B) \times C = \alpha(A \times C) + \beta(B \times C),$$

and

$$A \times B = -(B \times A).$$

Postulate 4 (Restricted norm compatibility). *For all $A \in V_R$,*

$$\|A \times R\| = \|A\| \|R\|.$$

Postulate 5 (Metric compatibility in V_R). *For all $A \in V_R$,*

$$\langle A \times R, A \rangle = 0.$$

Postulate 6 (Double-projection closure). *For all $A \in V_R$,*

$$(A \times R) \times R = \sigma R^2 A, \quad \sigma \in \{+1, -1\}.$$

Remark 1 (Algebraic scope). *Requiring $\|A \times B\| = \|A\| \|B\|$ for all A, B would force the setting toward normed composition algebra constraints. By contrast, the present framework imposes compatibility only relative to R . This restricted form is consistent with the intended role of R as a distinguished direction of comparison and avoids assuming a full composition algebra structure.*

Remark 2 (On σ). *The sign σ is a discrete degree of freedom attached to the double projection. When mapped to physical interpretation, it may play a spin-like role as a binary orientation class. In the internal formalism, however, it is simply a label distinguishing two closure types.*

2 Generated Directions

2.1 First projection: space of differences

Definition 1. *The first generated direction is*

$$U := \hat{\mathfrak{N}} \times R.$$

Since $\hat{\mathfrak{N}} \in V_R$, restricted norm compatibility gives

$$\|U\| = R.$$

Metric compatibility gives

$$\langle U, \hat{\mathfrak{N}} \rangle = 0, \quad \langle U, R \rangle = 0.$$

The mode U represents the first emergent informational degree of freedom: an arena of distinguishability.

2.2 Second projection: transport/orientation mode

Definition 2. *The second generated direction is*

$$V := U \times R.$$

Using the double-projection rule with $A = \hat{\mathfrak{N}}$,

$$V = (\hat{\mathfrak{N}} \times R) \times R = \sigma R^2 \hat{\mathfrak{N}}, \quad \|V\| = R^2.$$

The magnitude is independent of σ ; the direction encodes the orientation class.

2.3 Third projection: confinement/identity mode

We impose an orthogonal decomposition that separates the generated difference into a transport-like part and an orthogonal residual.

Postulate 7 (Orthogonal decomposition). *There exists $m \in V_R$ such that*

$$U = V + m, \quad \langle m, V \rangle = 0.$$

Then, by Pythagoras,

$$\|m\|^2 = \|U\|^2 - \|V\|^2 = R^2 - R^4,$$

and therefore

$$\|m\| = R\sqrt{1 - R^2}.$$

Remark 3 (Extrema). *The confinement/identity norm satisfies $\|m\| = 0$ at $R = 0$ and $R = 1$, and reaches its maximum at $R = 1/\sqrt{2}$.*

2.4 Fourth projection: interaction/adjustment mode

Definition 3. *The fourth generated direction is*

$$a := V \times R.$$

Since $V \in V_R$,

$$\|a\| = R^3.$$

This mode encodes the need for reconfiguration when relational modes interact: the seed of what becomes force-like or acceleration-like behavior after calibration.

2.5 Hierarchy summary

The generated hierarchy is

$$\|U\| = R, \quad \|V\| = R^2, \quad \|m\| = R\sqrt{1 - R^2}, \quad \|a\| = R^3.$$

3 Quadratic Harmonic Closure Depth

After calibration, using c and $\hbar$ as anchors, the norms are interpreted as frequency-like components of a single image frequency ω_I . The internal relation

$$\|U\|^2 = \|V\|^2 + \|m\|^2$$

becomes

$$\omega_I^2 = \omega_V^2 + \omega_m^2.$$

Postulate 8 (Binary quadratic closure). *A maximally stable family satisfies*

$$\frac{\omega_V^2}{\omega_I^2} = 2^{-\varepsilon}, \quad \varepsilon \in \mathbb{N}.$$

Equivalently,

$$\omega_V^2 : \omega_m^2 = 1 : (2^\varepsilon - 1).$$

This convention fixes ε as the quadratic harmonic depth. The associated Mersenne signature is

$$M_\varepsilon := 2^\varepsilon - 1 = \frac{\omega_m^2}{\omega_V^2}.$$

Remark 4 (Structural meaning of $\varepsilon = 4$). *This manuscript does not claim a theorem fixing $\varepsilon = 4$ as the only possible baryonic depth. It records a structural motivation. For a spin- $\frac{1}{2}$ system, the observable algebra is generated by the Pauli matrices, satisfying*

$$\sigma_i \sigma_j = \delta_{ij} I + i \epsilon_{ijk} \sigma_k, \quad (\mathbf{u} \cdot \boldsymbol{\sigma})^2 = I,$$

with $\mathbf{u}$ a unit vector. The latter identity is analogous to the return rule of double projection under the reading where R plays the role of a measurement axis.

The fundamental representation of $\mathfrak{su}(2)$ is two-dimensional over $\mathbb{C}$, hence four-dimensional over $\mathbb{R}$. In the present program, ε is interpreted as an encoding depth measuring the minimal independent real degrees needed to store a coherent non-commutative spin description alongside a reference state, a measurement axis, a non-commuting generator, and a binary orientation label. Under that reading, the baryonic assignment $\varepsilon = 4$ is the first depth compatible with this minimal real dimensionality. A complete selection rule remains open.

4 Proton-Scale Consistency Target

Let

$$\omega_p = \frac{m_p c^2}{\hbar}, \quad \bar{\lambda}_p = \frac{\hbar}{m_p c}.$$

For the baryonic instance $\varepsilon = 4$,

$$\omega_V = \omega_p 2^{-4/2} = \frac{\omega_p}{4}.$$

Define the associated projected length scale

$$R_V := \frac{c}{\omega_V} = 4\bar{\lambda}_p.$$

Using CODATA 2018 values,

$$R_V \approx 0.8412356 \text{ fm}.$$

Remark 5 (Falsifiability statement). *This numerical proximity to proton charge-radius determinations is presented as a consistency target of the $\varepsilon = 4$ instance, not as confirmation. The equality $R_V = r_p^{\text{charge}}$ is not an algebraic theorem of this manuscript; it requires an electromagnetic bridge hypothesis connecting the projected mode V with the proton electric form factor. If future consolidated determinations of r_p stabilize outside compatibility with that bridge, then the $\varepsilon = 4$ proton-radius correspondence is ruled out or must be reinterpreted.*

5 Program Notes Across Scales

Remark 6 (Status of this section). *The items below are not derived from the algebraic framework alone. They require additional interpretative steps and are presented as heuristic connections motivating further investigation. Each item should be treated as conditional and falsifiable.*

5.1 Atomic scale: valence heuristic

A proposed heuristic uses reduced ratios of valence occupations

$$(N_s : N_p) \longrightarrow (a : b)$$

and a weight

$$P = a + b,$$

where small- P structure correlates with inert or reactive patterns in periods 2 and 3. A derivation from the closure postulates remains open.

5.2 Cosmological scale: Hubble analogy

The internal scaling

$$R \longmapsto R^2$$

in the transport channel suggests a structural analogy with

$$V = H_0 D$$

under a suitable calibration. A dynamical model is required before any physical claim is made.

5.3 Gravitational scale: weak-field Poisson limit

One hypothesis introduces an effective relational density

$$\rho_{\text{rel}} := \kappa \omega_m^2 \rho_m$$

and proposes a Poisson-type limit

$$\nabla^2 \phi = 4\pi \alpha \kappa \omega_m^2 \rho_m, \quad G = \alpha \kappa \omega_m^2.$$

A key falsifiable target would be a small composition dependence, such as weak-equivalence-principle deviations, if $\kappa \omega_m^2$ varies with nuclear binding. Estimating the expected scale is an explicit next step.

6 Discussion and Open Questions

6.1 What this framework is and is not

This manuscript is a minimal generative formalism plus a calibration hypothesis. It does not yet provide:

1. a covariant dynamics;
2. a derivation of the calibration anchors;
3. a complete theorem fixing the admissible closure depths.

Its value lies in conceptual economy, explicit separation of postulates from derived relations, and a compact list of falsifiable targets.

6.2 Open questions

1. Can the return rule and restricted norm compatibility be represented as a known algebraic structure, for example a Lie-type representation, without contradiction?
2. Can the quadratic harmonic depth ε be derived as a counting theorem of independent relational constraints, beyond the structural motivation in the spin- $\frac{1}{2}$ case?
3. What is the minimal covariant dynamics compatible with the generative operator $\times$?
4. What empirical signature most cleanly distinguishes this program from reparameterizations of known physics, for example composition-dependent weak-equivalence-principle tests?
5. How should the electromagnetic bridge between the projected mode V and observed charge structure be derived from an effective current?

Conclusion

We constructed a minimal relational-generative algebra in which new degrees of freedom arise by successive projections. A binary orientation σ emerges through double projection. An orthogonal decomposition yields a confinement-like mode with norm

$$R\sqrt{1 - R^2}$$

and a natural hierarchy of scalings:

$$U, \quad V, \quad m, \quad a.$$

A binary quadratic closure hypothesis discretizes stable families by depth ε , and the baryonic choice $\varepsilon = 4$ receives a structural rationale from the non-commutative structure of spin and the real dimension of the fundamental representation of $\mathfrak{su}(2)$. As a concrete consistency check, the proton Compton frequency combined with the $\varepsilon = 4$ squared-channel ratio

$$1 : 15$$

yields a projected scale

$$R_V = 4\bar{\lambda}_p \approx 0.8412 \text{ fm},$$

close to the experimentally determined proton charge radius under the additional electromagnetic bridge hypothesis.

Future work includes:

1. classification of admissible realizations satisfying restricted norm and metric compatibility;
2. a genuine selection rule for the harmonic depth ε ;
3. a dynamical extension of the generative operator;
4. derivation of the electromagnetic bridge from an effective current;
5. connection with gravitational, atomic, and cosmological limits without importing their mathematical structures prematurely.

Epilogue

In a point without references, where everything continues without “happening”, perhaps the present is not something we inhabit, but something we are.

And the dimensions we discover are not containers we move through, but spaces we create each time information needs a place to be.

A Summary of Objects and Their Interpretations

Object	Internal meaning	Possible physical analog / interpretation
$\hat{\mathfrak{N}}$	Normalized reference state	Unison present; global phase; vacuum-like reference
R	Differentiation marker	Measurement axis; direction of comparison
$U = \hat{\mathfrak{N}} \times R$	First projection: space of differences	Non-commuting generator; arena of distinguishability
σ	Binary orientation in double projection	Spin orientation; helicity; particle-antiparticle class
$V = U \times R$	Second projection: transport mode	Relational velocity; momentum-like channel
m	Third projection: confinement/identity mode	Mass/inertia proxy; self-identity channel
$a = V \times R$	Fourth projection: interaction mode	Force-like or adjustment channel
ε	Quadratic harmonic depth	Number of independent binary closure distinctions
$M_\varepsilon = 2^\varepsilon - 1$	Mersenne signature	Confined-to-projected squared ratio

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